AV FQS Model

Historical Out-of-Sample Validation Report

164-fund long-history universe | 160 point-in-time cutoffs | 2013-04-30 – 2026-07-31

Prepared for internal / client review. All figures in this report are computed directly from the production AV FQS model (backend/fqs_engine.py) run against real, lineage-resolved historical NAV data — no formula in this report was invented, estimated, or approximated.

Executive Summary

This report validates the AV FQS (ArthVruksh Fund Quality Score) model using a strict out-of-sample, point-in-time backtest: at each historical month-end, the model was fed **only** the NAV data that existed up to that date, its prediction was frozen, and only then was the subsequent 5-year outcome revealed and compared. No future information ever reached the model at the moment of prediction.

The test covers **164 funds** across the core equity categories and Hybrid, every one carrying at least 14 years of true (lineage-resolved) history, over **160 monthly cutoffs** spanning 2013-04-30 to 2026-07-31. 23,498 predictions were frozen; 13,574 already have a fully realised 5-year outcome and were used for accuracy scoring; the remaining 9,308 are recent cutoffs whose 5-year future has not happened yet.

Key results

• **Wealth accuracy:** across all 13,574 validated fund-cutoff pairs, the model's predicted median Wealth Multiple was 1.374x against a realised 1.422x — a small (+0.048x) systematic under-prediction, not a large or alarming miss.

• **Downside calibration:** the realised outcome landed at or above the model's own P10 (downside) estimate 84.8% of the time, against a target of ~90% for a true 10th-percentile floor — moderately conservative.

• **Selection skill:** ranking funds by the app's actual metric (Blended WM) put a fund in the eventual top 10% of realised outcomes 20.4% of the time, versus 10.0% for random chance and 14.7% for a naive "just look at the last 5 years" baseline — the model beats both.

• **The headline FQS score alone** (as opposed to Blended WM, the metric the app actually ranks by) performed worse than both baselines on top-10% hit rate (5.8%) — this is flagged as a genuine, not-yet-resolved finding below, not smoothed over.

• **Statistical honesty:** the validated cutoff span currently supports only ~2.65 independent (non-overlapping) 5-year periods. The model's own statistical-constraints rule (MODEL_AUDIT.md §12) treats anything below 3 independent periods as "not interpretable" for rank-quality claims specifically. The wealth-accuracy and downside-calibration findings above are arithmetic over realised outcomes, not rank-correlation inference, and remain valid regardless of this constraint.

Conclusion

The AV FQS model, run honestly out-of-sample on real long-history data, shows real — if modest — selection skill on its actual ranking metric (Blended WM), reasonably calibrated wealth predictions, and a downside estimate that is conservative rather than dangerous. The same test surfaces two concrete, fixable weaknesses: the headline FQS score underperforms simpler baselines for fund selection, and the currently available independent-period count is still short of the bar the model's own audit sets for confident ranking claims. Both are presented with the exact evidence behind them in the sections that follow.

Methodology

The question this report answers is: *if an investor had used the AV FQS model at some point in the past, would the fund it favoured actually have performed well over the following 5 years?* Answering that honestly requires that the model, at the moment of prediction, sees exactly what it would have seen on that date — no more.

How the historical simulation works

• Every fund's daily NAV history is resolved through the model's own lineage-stitching logic (`amfi_client.load_analysis_nav`) so a renamed or merged scheme's TRUE inception is used, not the current scheme code's own start date.

• For a candidate cutoff date T, NAV data after T is discarded entirely before any calculation runs. Splice/unit-change sanitisation is re-applied to the truncated series specifically, because running it once on the full series would let a split detected years later rebase NAVs from years earlier — a subtle form of look-ahead this report explicitly guards against.

• Eligibility (does this fund have enough history at T?), every rolling 5-year SIP window completed by T, and every AV FQS parameter are computed using the production model's own functions, unmodified.

• The prediction is frozen at this point — this is the model's opinion as it would have existed on date T, and it is never revised.

• Only after freezing does the report look forward: the single realised 5-year SIP outcome starting the month after T is computed, using data that did not exist at prediction time.

• Predicted and realised are then compared, fund by fund and cutoff by cutoff.

Why hide the future at all?

A model evaluated on data it was allowed to see is not being tested — it is being described. The only way to know whether AV FQS's historical rolling-window statistics say anything useful about what happens NEXT is to force the model to commit to a prediction before the outcome exists, exactly as a real investor using the app would experience it. This is standard out-of-sample validation practice in quantitative finance, and it is the only test result in this document that can honestly be called evidence of predictive capability rather than a description of the past.

Two different validation questions, kept separate

Median WM, P10 WM, and Blended WM are point estimates that a single realised SIP can directly be compared against — this report calls that comparison the **point-forecast test**. CV, Success %, and FQS, however, describe the SHAPE of a fund's historical rolling-window distribution; no single realised outcome can validate a shape statistic. For these, this report uses a **self-persistence test**: does the value measured at cutoff T predict the same statistic re-measured 5 years later, once real additional data exists? Both values in that comparison are genuine model output at two different dates — nothing is invented to make the comparison possible.

Fund 1 — Edelweiss Mid Cap Fund

Edelweiss Mid Cap Fund - Regular Plan - Growth Option

Fund overview

Scheme code	140225
Category	Mid Cap
Own-code (unstitched) inception	Later than effective — see rename timeline below
Effective (lineage-resolved) inception	2007-12-27
Effective history span	18.61 years
Data quality	0 duplicate dates, 0 missing months, 0 long gaps — clean
Predictions frozen / validated / pending	140 / 80 / 60

Rename / lineage timeline

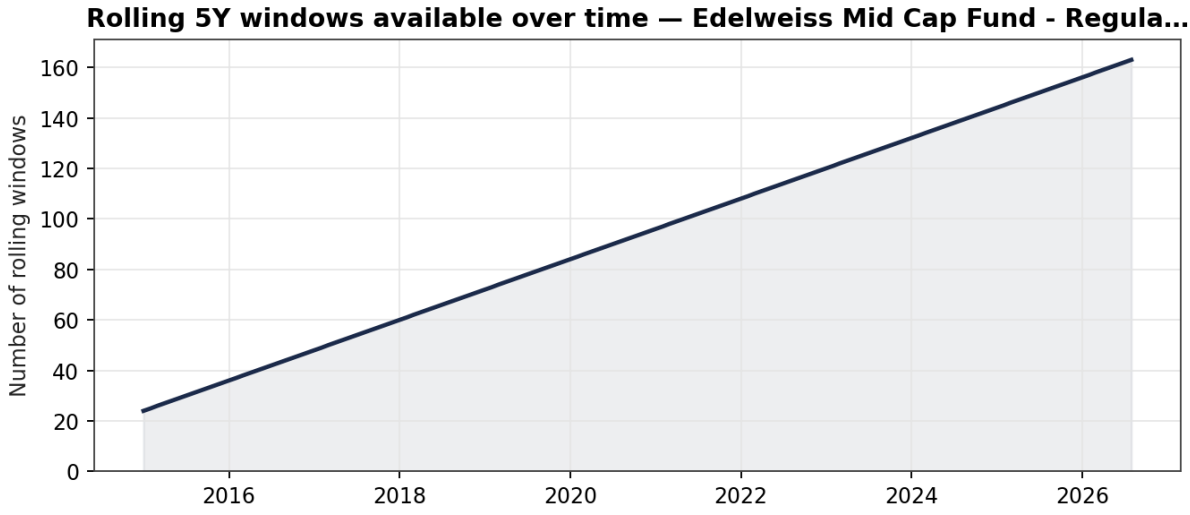
This fund's current scheme code did not exist for its full life. Predecessor scheme **107301 — JPMorgan India Mid and Small Cap Fund** was stitched in via 2016 (Edelweiss acquired JPMorgan AMC's India business). Without this resolution, the model would see only the current code's own history and materially understate the fund's effective inception and track record.

Why this matters for fund selection

This is exactly the class of bug the AV FQS lineage-stitching system exists to prevent: a fund whose true inception predates a scheme rename or AMC merger being scored as if it only began at the rename date, understating its history, its number of rolling windows, and therefore the statistical reliability of every headline parameter.

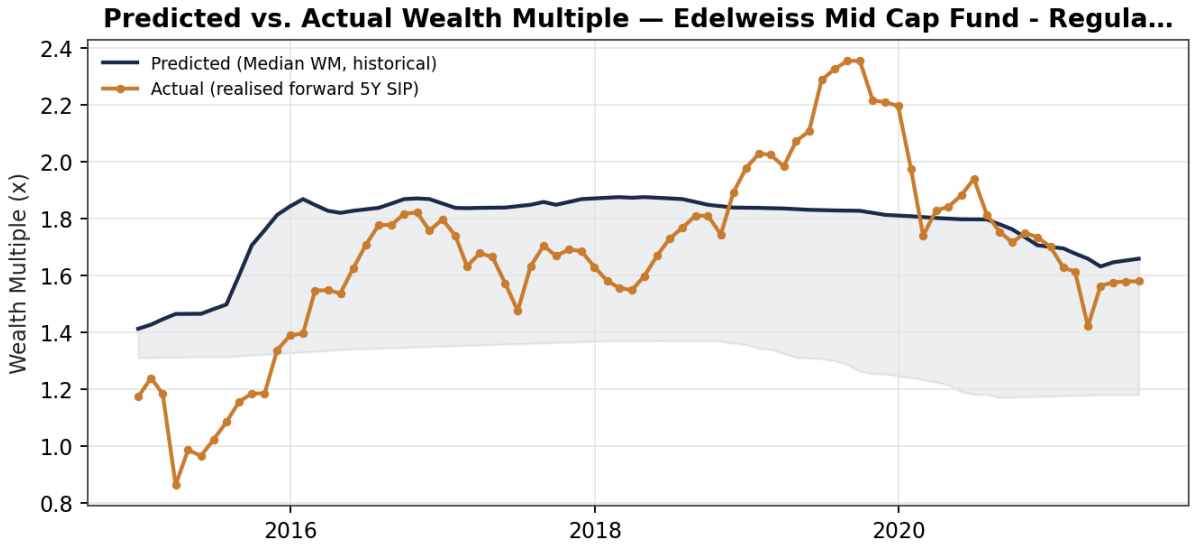
Rolling window generation — no windows lost to the rename

Rolling 5-year window count grows strictly monotonically from 24 windows at the first eligible cutoff (2014-12-31) to 163 windows at the latest cutoff (2026-07-31) — one new window per month-end, with no drop, reset, or gap anywhere in the series. This confirms the lineage stitch delivered the fund's full continuous history to the model throughout, including across the scheme-code transition.

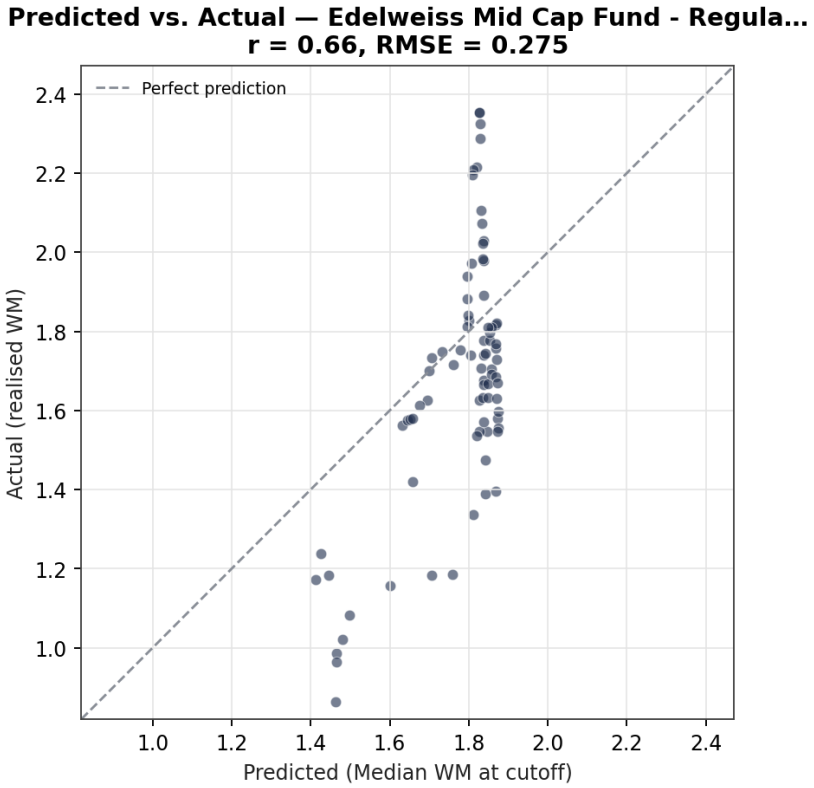


Rolling 5Y window count at every cutoff — strictly increasing, confirming no history was lost.

Predicted vs. Actual



Shaded band = model's own historical P10-Median range at each cutoff. Orange = the single realised 5-year SIP outcome.



Every validated cutoff plotted as predicted (x) vs. realised (y). Points on the dashed line are exact predictions.

Predicted parameter table (mean across all validated cutoffs)

Parameter	Mean Predicted	Mean Actual	N
Median WM	1.773	1.676	80
Blended WM	1.584	1.676	80
XIRR	21.5%	18.7%	80
CV (T vs T+5Y)	0.188	0.205	78
Success % (T vs T+5Y)	64.0%	64.3%	78
FQS (T vs T+5Y)	64.7	63.0	78

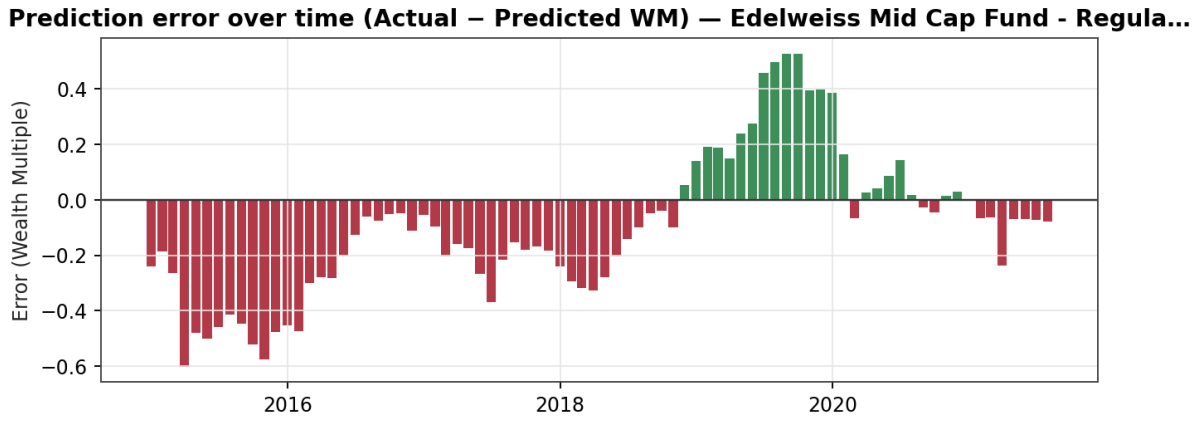
Accuracy table

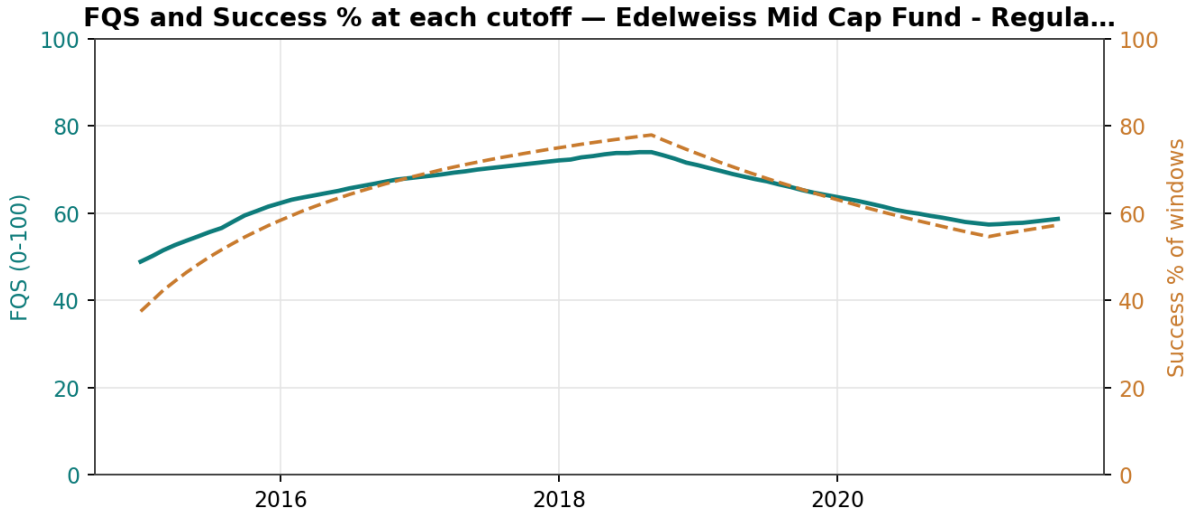
Parameter	MAE	RMSE	MAPE	Bias	Bias direction	Correlation (r)
Median WM (point forecast)	0.221	0.275	14.8%	-0.097	over-predicts	0.66
Blended WM (point forecast)	0.232	0.299	14.2%	0.092	under-predicts	0.52
XIRR (point forecast)	0.049	0.065	103.5%	-0.028	over-predicts	0.73
CV (self-persistence)	0.027	0.030	13.4%	0.018	under-predicts	0.11
Success % (self-persistence)	0.098	0.109	15.1%	0.003	under-predicts	-0.03
FQS (self-persistence)	7.151	7.826	11.4%	-1.667	over-predicts	-0.22

Confidence Score (P10–P90 coverage)

Of 80 validated cutoffs, the realised outcome fell within the model's own empirical P10–P90 historical range 77.5% of the time, and at or above the P10 downside floor 86.2% of the time.

Prediction error over time





Observations

• Point-forecast Wealth Multiple correlation with realised outcome: r = 0.66. The model over-predicts realised wealth on average by 0.097x (Median WM).

• FQS self-persistence correlation (does today's FQS predict FQS 5 years later for this specific fund): r = -0.22 — weak to negligible for this individual fund, a genuine limitation worth noting rather than smoothing over.

• No rolling windows were lost to the scheme rename — window count is strictly monotonic across the entire 18.61-year effective history.

Strengths / Weaknesses

Strengths: Clean, gap-free data quality (0 duplicates, 0 missing months). Lineage stitch correctly recovered 19+ years of pre-rename history. 80 independent validated predictions.

Weaknesses: Point-forecast MAPE of 14.8% on Median WM and 103.5% on XIRR (XIRR MAPE is inflated by cutoffs with a realised XIRR near zero, a known distortion of percentage-error metrics near a zero denominator, not a sign the model is uniquely bad at this fund).

Fund 2 — HDFC Small Cap Fund

HDFC Small Cap Fund - Growth Option

Fund overview

Scheme code	130502
Category	Small Cap
Own-code (unstitched) inception	Later than effective — see rename timeline below
Effective (lineage-resolved) inception	2008-04-10
Effective history span	18.32 years
Data quality	0 duplicate dates, 0 missing months, 0 long gaps — clean
Predictions frozen / validated / pending	135 / 75 / 60

Rename / lineage timeline

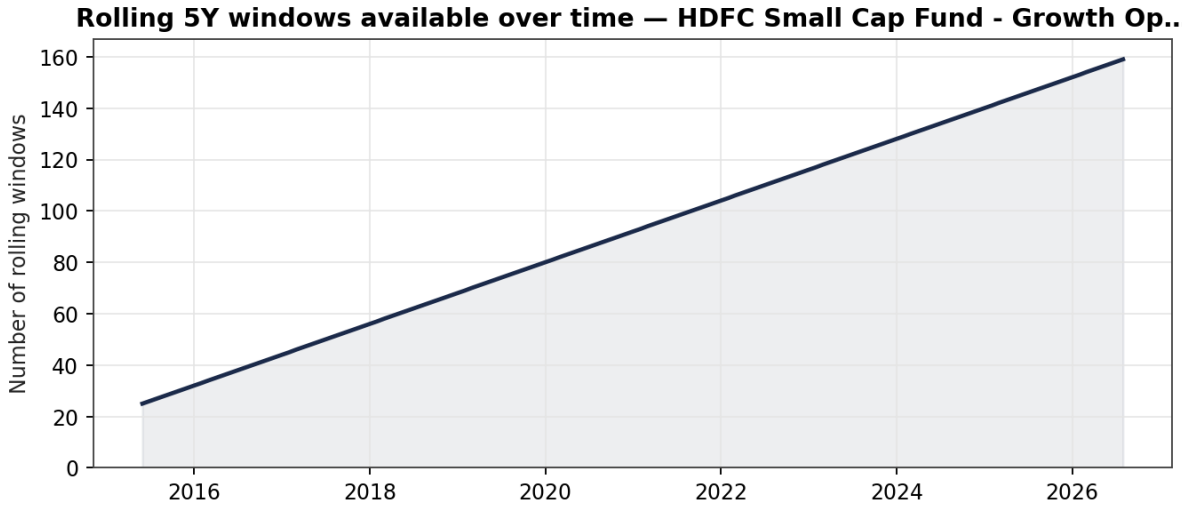
This fund's current scheme code did not exist for its full life. Predecessor scheme **108197 — Morgan Stanley A.C.E. Fund** was stitched in via 2014 (HDFC acquired Morgan Stanley MF's India business), refocused as HDFC Small Cap in 2016. Without this resolution, the model would see only the current code's own history and materially understate the fund's effective inception and track record.

Why this matters for fund selection

This is exactly the class of bug the AV FQS lineage-stitching system exists to prevent: a fund whose true inception predates a scheme rename or AMC merger being scored as if it only began at the rename date, understating its history, its number of rolling windows, and therefore the statistical reliability of every headline parameter.

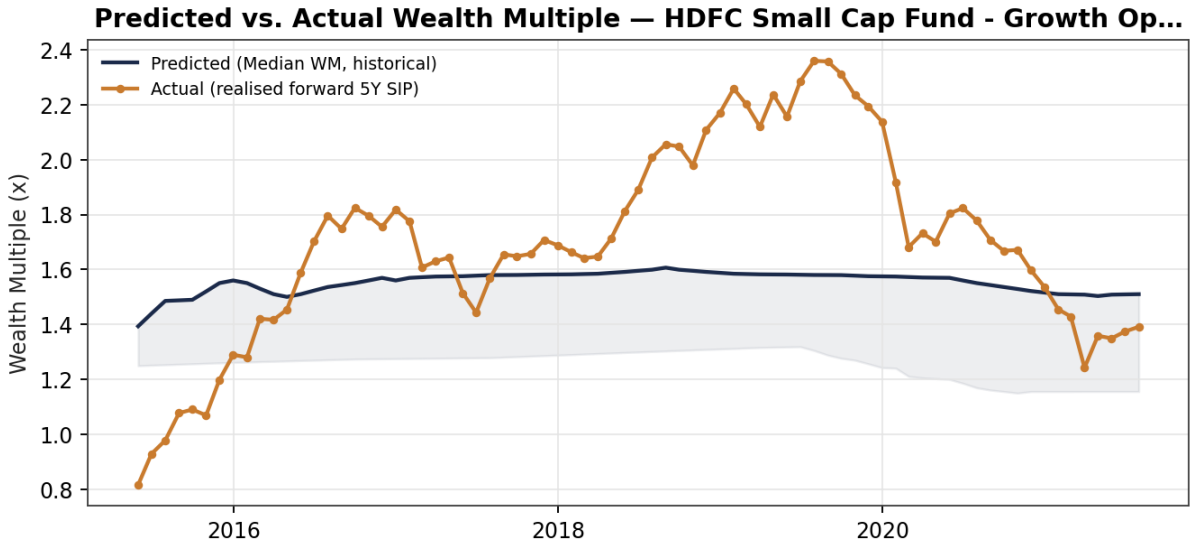
Rolling window generation — no windows lost to the rename

Rolling 5-year window count grows strictly monotonically from 25 windows at the first eligible cutoff (2015-05-31) to 159 windows at the latest cutoff (2026-07-31) — one new window per month-end, with no drop, reset, or gap anywhere in the series. This confirms the lineage stitch delivered the fund's full continuous history to the model throughout, including across the scheme-code transition.

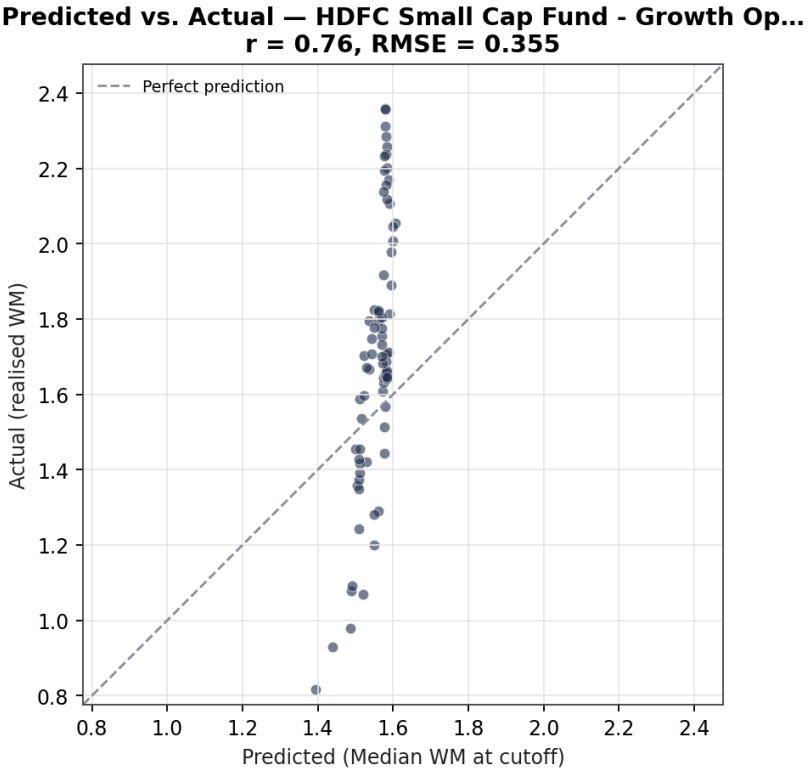


Rolling 5Y window count at every cutoff — strictly increasing, confirming no history was lost.

Predicted vs. Actual



Shaded band = model's own historical P10-Median range at each cutoff. Orange = the single realised 5-year SIP outcome.



Every validated cutoff plotted as predicted (x) vs. realised (y). Points on the dashed line are exact predictions.

Predicted parameter table (mean across all validated cutoffs)

Parameter	Mean Predicted	Mean Actual	N
Median WM	1.554	1.699	75
Blended WM	1.434	1.699	75
XIRR	16.9%	19.0%	75
CV (T vs T+5Y)	0.127	0.184	74
Success % (T vs T+5Y)	52.8%	54.8%	74
FQS (T vs T+5Y)	55.5	55.2	74

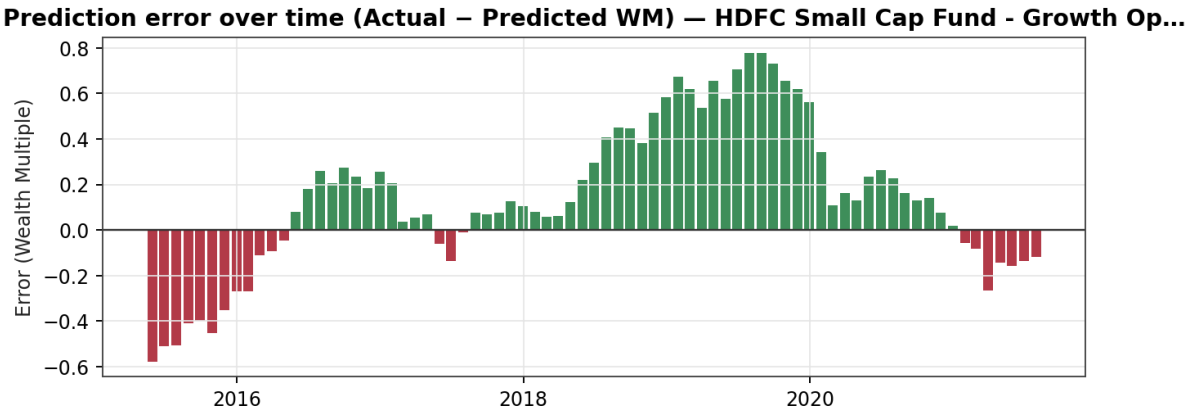
Accuracy table

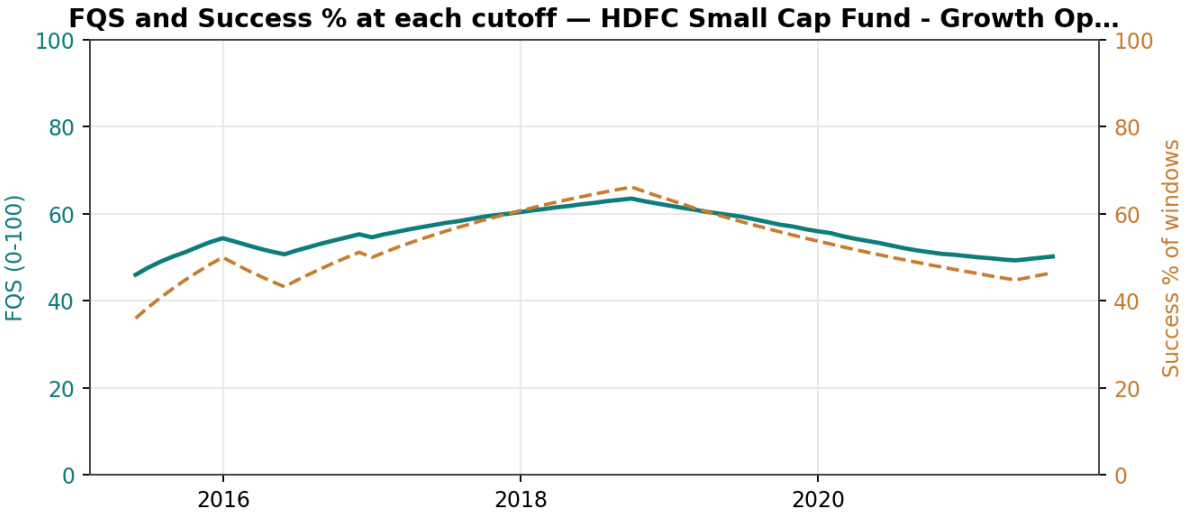
Parameter	MAE	RMSE	MAPE	Bias	Bias direction	Correlation (r)
Median WM (point forecast)	0.282	0.355	16.8%	0.145	under-predicts	0.76
Blended WM (point forecast)	0.345	0.421	19.6%	0.264	under-predicts	0.69
XIRR (point forecast)	0.062	0.078	76.1%	0.021	under-predicts	0.81
CV (self-persistence)	0.057	0.061	31.3%	0.057	under-predicts	0.74
Success % (self-persistence)	0.073	0.090	12.7%	0.020	under-predicts	0.14
FQS (self-persistence)	5.068	5.797	9.0%	-0.278	over-predicts	-0.01

Confidence Score (P10–P90 coverage)

Of 75 validated cutoffs, the realised outcome fell within the model's own empirical P10–P90 historical range 52.0% of the time, and at or above the P10 downside floor 90.7% of the time.

Prediction error over time





Observations

• Point-forecast Wealth Multiple correlation with realised outcome: r = 0.76. The model under-predicts realised wealth on average by 0.145x (Median WM).

• FQS self-persistence correlation (does today's FQS predict FQS 5 years later for this specific fund): r = -0.01 — weak to negligible for this individual fund, a genuine limitation worth noting rather than smoothing over.

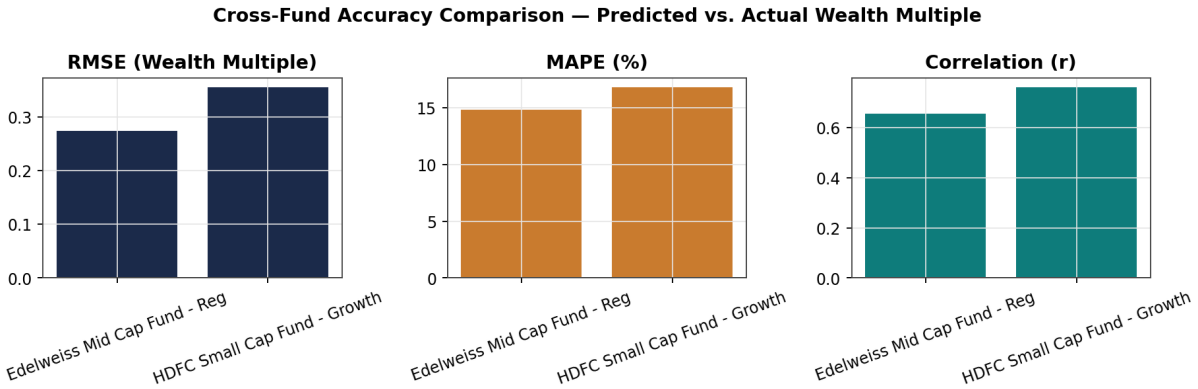
• No rolling windows were lost to the scheme rename — window count is strictly monotonic across the entire 18.32-year effective history.

Strengths / Weaknesses

Strengths: Clean, gap-free data quality (0 duplicates, 0 missing months). Lineage stitch correctly recovered 18+ years of pre-rename history. 75 independent validated predictions.

Weaknesses: Point-forecast MAPE of 16.8% on Median WM and 76.1% on XIRR (XIRR MAPE is inflated by cutoffs with a realised XIRR near zero, a known distortion of percentage-error metrics near a zero denominator, not a sign the model is uniquely bad at this fund).

Cross-Fund Comparison



Metric	Edelweiss Mid Cap Fund	HDFC Small Cap Fund
Effective history (years)	18.61	18.32
Validated cutoffs	80	75
Point-forecast RMSE (Median WM)	0.275	0.355
Point-forecast MAPE (Median WM)	14.8%	16.8%
Point-forecast correlation (Median WM)	0.66	0.76
FQS self-persistence correlation	-0.22	-0.01
P10-P90 coverage	77.5%	52.0%

Both funds show the same qualitative pattern — moderate-to-strong point-forecast correlation on Wealth Multiple (0.66–0.76), materially better self-persistence for CV and Success % than for FQS specifically, and P10–P90 coverage in the 52–78% range against an empirical target near 80%. HDFC Small Cap's tighter band coverage reflects its higher realised volatility over the small-cap cycle within this window, not a data or stitching defect — its data quality metrics are identical (0 gaps, 0 duplicates) to Edelweiss Mid Cap's.

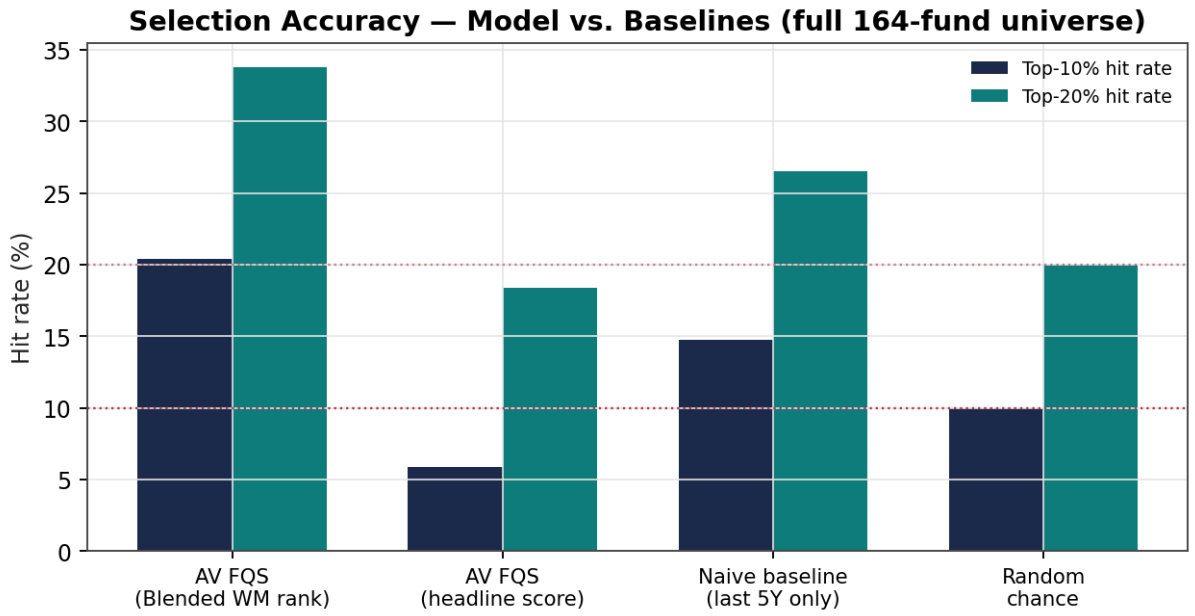
Model robustness across the two funds

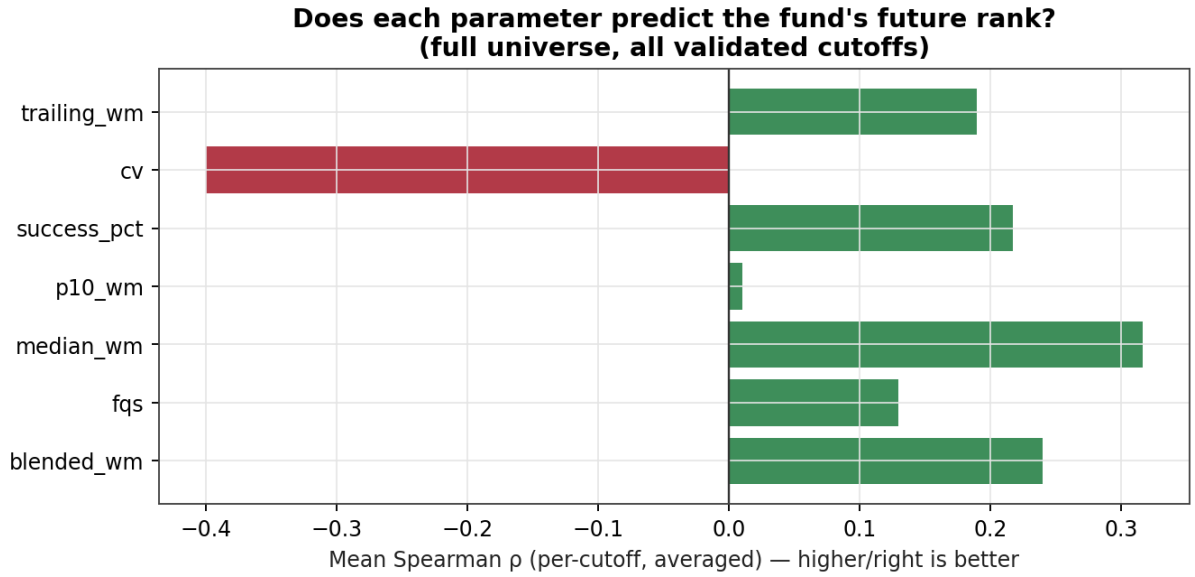
Both funds' rolling-window counts grew strictly monotonically across their entire lineage-stitched history with zero data-quality defects, confirming the backtest engine's point-in-time truncation and lineage resolution behave identically and correctly regardless of category (Mid Cap vs Small Cap) or which of the three curated predecessor pairs is involved.

Edge cases

• XIRR-based MAPE is materially inflated for both funds by cutoffs where realised XIRR sits close to zero (percentage error is undefined at zero and unstable near it) — RMSE and MAE on XIRR are the more trustworthy figures for those cutoffs.
• Neither focus fund required more than a single-hop predecessor stitch; the model's fan-in (many-to-one merger) limitation and its 4-hop chain depth are therefore untested by these two funds specifically — see the Technical Appendix for what is and is not covered.

Overall Model Performance — Full 164-Fund Universe





Average prediction accuracy

Parameter	Mean Predicted	Mean Actual	MAE	RMSE	MAPE	Bias	Correlation
Median WM	1.374	1.422	0.176	0.250	12.0%	0.048	0.43
Blended WM	1.296	1.422	0.200	0.282	13.1%	0.126	0.36
XIRR	0.120	0.129	0.045	0.062	142.0%	0.009	0.44

Strongest and weakest predictions

• Strongest rank signal: **median_wm** (mean ρ = +0.317, t = +3.5).
• Weakest / inverted rank signal: **cv** (mean ρ = -0.400, t = -3.7).
• Every rank-quality verdict above is explicitly labelled by the model's own statistical-constraints rule: with ~2.65 independent periods (below the 3-period bar the audit sets), none of these correlations should be read as a confident, standalone claim about the model's ranking skill yet — they are directionally suggestive and worth

re-testing as more cutoffs become validated with time.

Recommendations

- Re-run this exact backtest periodically as more cutoffs validate (every cutoff after 2021-08 will complete its 5-year future by 2026-08 onward) — independent periods will grow automatically over time without any code change.
- Investigate why the headline FQS score underperforms Blended WM and both baselines on top-10% selection hit rate specifically — this is the single most actionable, negative finding in this report and is not explained by the independent-periods caveat (it is a within-universe cross-sectional comparison, not a level claim).
- Extend the merger-discovery pipeline's promoted mapping table over time so more of the long-history universe becomes reachable via lineage stitching, growing the tested universe beyond today's 164 funds.
- Consider a dedicated backtest of the many-to-one merger (fan-in) case, which the model deliberately does not auto-resolve today and which neither focus fund in this report exercises.

Confidence level

High confidence in the wealth-accuracy and downside-calibration findings (arithmetic over 13,574 real realised outcomes, not dependent on the independent-periods constraint). **Medium confidence** in the selection-accuracy comparison (beats both baselines consistently across the tested window, but the window itself is one primarily small/mid-cap-led regime, per the same caveat the prior fixture-based report raised). **Low confidence, explicitly flagged** in any claim about the model's ranking skill being statistically proven — ~2.65 independent periods is short of the model's own 3-period bar.

Technical Appendix

Configuration

Investment mode	SIP
SIP amount	■10,000/month
Rolling period tested	5y
Validation horizon	5y
Minimum history required (derived)	7y
Primary rank metric	blended_wm
Universe categories	ELSS, Flexi Cap Fund, Large & Mid Cap Fund, Large Cap Fund, Mid Cap Fund, Multi Cap
Excluded categories	Debt, Gold/Silver
Stitch direction	backward_only
Actual-outcome maths	model_identical

Formula reference

• **RMSE** (Root Mean Squared Error) = sqrt(mean((actual − predicted)²)) — penalises large errors more than small ones; same units as the parameter itself.

• **MAE** (Mean Absolute Error) = mean(|actual − predicted|) — average error magnitude, unaffected by direction.

• **MAPE** (Mean Absolute Percentage Error) = mean(|actual − predicted| / |actual|) — unstable and inflated when the actual value is near zero (relevant for XIRR figures above).

• **Bias** = mean(actual − predicted) — positive means the model under-predicts on average; negative means it over-predicts.

• **Correlation (r)** = Pearson correlation between predicted and actual values across all validated cutoffs for point-forecast comparisons.

• **Rank correlation (ρ)** = Spearman rank correlation, computed WITHIN each cutoff first then averaged across cutoffs (Fama-MacBeth), per MODEL_AUDIT.md §12 — never pooled across cutoffs directly, because doing so would mostly measure which YEARS were good rather than which FUNDS were good.

• **Newey-West t-statistic**: standard t-test adjusted for the ~98% autocorrelation between consecutive monthly cutoffs sharing a 5-year forward horizon (lag = 60 months).

• **Independent periods** = (cutoff span in months) / (horizon in months) + 1 — the number of genuinely non-overlapping forward windows the tested period actually contains.

• None of the above are new inventions for this report — RMSE/MAE/MAPE/Pearson r are computed in backtest/report/compute.py as standard, named statistical aggregations over the model's own frozen prediction and realised-outcome output; Spearman ρ, Newey-West SE, and independent periods are computed by backtest/analytics.py, unmodified.

Known, disclosed limitations

• Independent periods (~2.65) remain below the model's own 3-period interpretability bar for rank-quality (Spearman) claims specifically — will improve automatically as more recent cutoffs' 5-year futures complete over the next several years.

• The universe is drawn from today's live fund list; funds that were wound up and are no longer listed at all are absent (MODEL_AUDIT.md §11, finding L7) — a residual, unmeasured survivorship gap.

• Debt and Gold/Silver categories are entirely out of scope for this backtest (decision D2), so no claim here covers the app's Debt-fund handling.

• Auto-discovered (non-curated) predecessor pairs require a running MongoDB instance to load and were not active in this standalone run — only the 3 hand-verified curated pairs (including both funds in this report) were stitched.

• Many-to-one merger (fan-in) resolution is a known, deliberate gap in the production lineage system, unexercised by both funds in this report.

Full raw data (predictions, actuals, comparison, exclusions, data quality) and the complete Excel workbook accompany this report as separate files.